Run report: k = 4 exponent ladder (ecbench), 2026-10-15T04:57Z

Descriptive. What ran, and what it recorded. Interpretation is in the study's `README.md` and `REPORT.pdf`.

- **Study:** `research/gaudry_quartic_d4_exponent_20261015`, protocol `PROTOCOL.md` (registered at `08722a2b7`; amendments 1–3). Code `0bea75038`; `ecbench` sha256 `8627cd29…0aa3`.
- **Specs:** `specs/SPEC-<job>.json`, one per (arm, p, target); generated by `make_specs.sh`. Queue: `jobs.txt` (amendment 1 order), run by `run_queue.sh`, `run_queue2.sh`, `run_queue3.sh`; log `sessions/queue.log`.
- **Host:** Claude Code cloud container, Intel Xeon @ 2.80 GHz, 4 vCPU at launch, 3 (CPUs 0–2) from about 16:30Z; rustc 1.97.0. Every session ran with `--allow-busy`: isolation earned L0 (229 records) or L1 (11). `S` is a count (`priced_by = counted`) and does not depend on wall time; wall time below is descriptive only.
- **Workloads:** `fp4_torsion(p, seed 1, max_cofactor 16)`, `p ∈ {29, 37, 53, 61, 101, 149, 197, 293, 401, 509, 701, 1009}`; target seed `2026101500000 + 10p + t` (IC specs, one target each), `2026101590000 + p` (rho-only, 16 targets); measurement seed `202610150`.
- **Arms:** `rho-neg` (`rho.negation`); `ic-d4-w` (`ic.pipeline, fghr-y-line:zero=1, gaudry-d4:macaulay=26,weighted=1,chains=1,sparsest=1,replay=1,charge_unsolved=1, linalg=wiedemann`); `ic-d4-lp2` and `ic-d4-lp1` (the same with `large_primes=2` and 1).

Runs requested and completed

72 jobs were planned. **59 ran and completed**; every record in them verified (`[d]G = Q`): 27 `ic-d4-w`, 9 `ic-d4-lp2`, 12 `ic-d4-lp1`, and 240 `rho-neg` records (219 distinct; the IC specs' paired rho runs repeat the same seed when two arms share a target). **13 did not start before the 16 h stop** (`sessions/NOT_RUN.txt`, `NOT_RUN3.txt`): `main-p{29,37,53,61,197}-t2` and `lp2-p{29,37,53,61,149,197,293}-t1`. 20 jobs failed at launch at about 16:28Z ("CPU n is outside this process's cpuset"; logs `sessions/*.launch-failed.log`, no computation) and were relaunched (amendment 3); 7 of them completed, 13 are the ones above. From 16:31Z to 20:34Z the relaunched jobs ran pinned to one CPU (wall only).

Erratum, 2026-10-19. The counts above are wrong: **60** jobs ran and **12** did not start before the stop, not 59 and 13 (48 IC and 12 rho-only sessions on disk; the 20 relaunched jobs were 8 completed and 12 not started, not 7 and 13). The 12 were run on 2026-10-19; see `reports/run-20261019T1612Z/`. The original text is left as written. The table below lists the 48 index-calculus sessions of the first 60 jobs, and the rho-only runs are counted in `runs.tsv`.

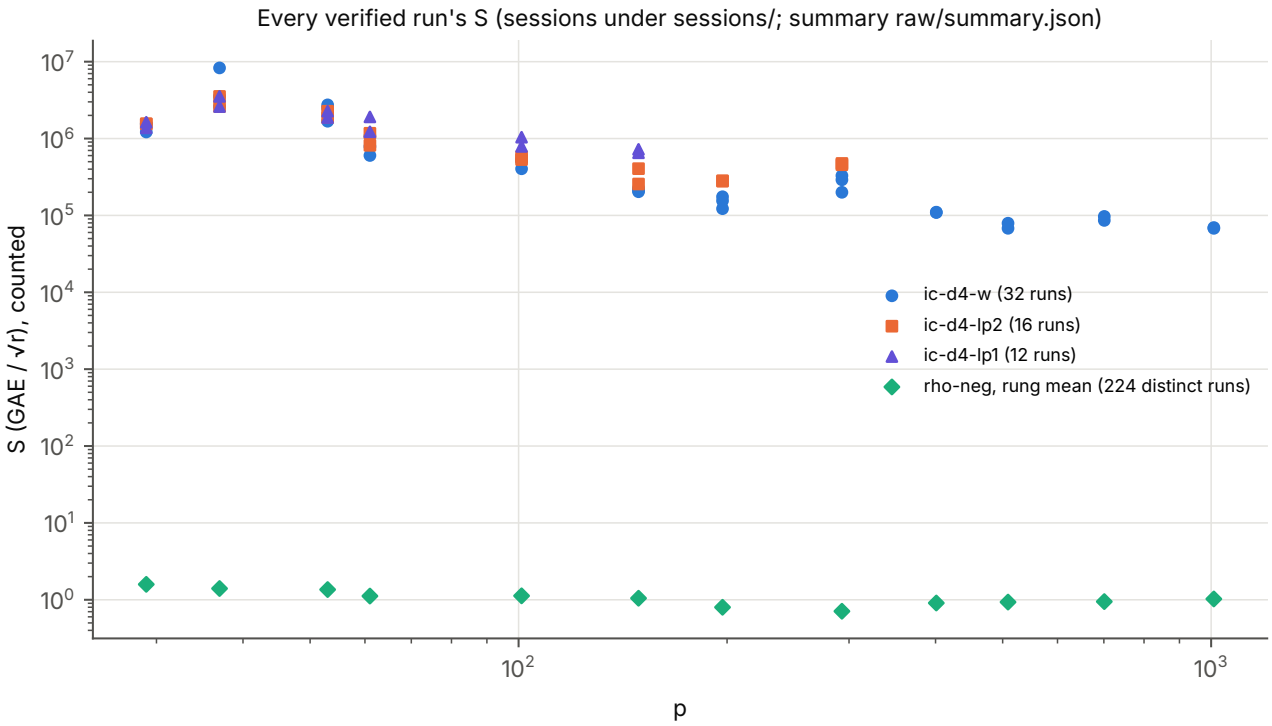


Figure 1: Every verified run's `S` by arm against `p` (48 IC runs; rho as rung means of the 219 distinct runs). Sessions under `sessions/`; summary `raw/summary.json`.

Index-calculus runs (measured; S in GAE/√r, trials counted)

job	session	arm	p	status	S	trials	wall h
lp1-p101-t0	ECBS1h1069073f92d6	ic-d4-lp1	101	verified	789502	3187	0.38
lp1-p101-t1	ECBS1h196d20e6ae07	ic-d4-lp1	101	verified	1040474	4152	1.63
lp1-p149-t0	ECBS1h9c5487dfb184	ic-d4-lp1	149	verified	723818	12465	4.92
lp1-p149-t1	ECBS1h75bb2698ae80	ic-d4-lp1	149	verified	652380	11218	4.72

lp1-p29-t0	ECBS1h2cb9ccf6ab7a	ic-d4-lp1	29	verified	1629980	583	0.06
lp1-p29-t1	ECBS1h8de8328b7051	ic-d4-lp1	29	verified	1376603	458	0.18
lp1-p37-t0	ECBS1hff5ab0a44a7a	ic-d4-lp1	37	verified	3531625	2838	0.33
lp1-p37-t1	ECBS1hcd46e190cc3f	ic-d4-lp1	37	verified	2604753	2117	0.78
lp1-p53-t0	ECBS1hfacb6bb78b3e	ic-d4-lp1	53	verified	1873288	3091	0.36
lp1-p53-t1	ECBS1ha5ee1295e3f2	ic-d4-lp1	53	verified	2265837	3691	1.37
lp1-p61-t0	ECBS1h36fcf7375005	ic-d4-lp1	61	verified	1223435	1937	0.21
lp1-p61-t1	ECBS1hc5f03470ecd0	ic-d4-lp1	61	verified	1910709	2899	0.43
lp2-p101-t0	ECBS1h99f39559aaf1	ic-d4-lp2	101	verified	532629	2148	0.24
lp2-p101-t1	ECBS1hcb323db9312b	ic-d4-lp2	101	verified	580541	2318	0.33
lp2-p149-t0	ECBS1hb0c914fd1eff	ic-d4-lp2	149	verified	256239	4405	0.5
lp2-p197-t0	ECBS1h72c6c3c2e245	ic-d4-lp2	197	verified	278749	8228	0.94
lp2-p29-t0	ECBS1ha6d51db65c5c	ic-d4-lp2	29	verified	1549971	552	0.07
lp2-p293-t0	ECBS1h5fe5a6d9d5c8	ic-d4-lp2	293	verified	474327	19807	2.87
lp2-p37-t0	ECBS1ha0f90a001534	ic-d4-lp2	37	verified	3531625	2838	0.33
lp2-p53-t0	ECBS1hfb8b21de58d4	ic-d4-lp2	53	verified	1873288	3091	0.36
lp2-p61-t0	ECBS1h64cbd25116d1	ic-d4-lp2	61	verified	816156	1293	0.14
main-p1009-t0	ECBS1h2230dd3464c1	ic-d4-w	1009	verified	67902	20706	4.18
main-p1009-t1	ECBS1h42ae79ebb072	ic-d4-w	1009	verified	69527	21309	4.28
main-p101-t0	ECBS1h42b956952e13	ic-d4-w	101	verified	404342	1627	0.19
main-p101-t1	ECBS1hc254ec885a58	ic-d4-w	101	verified	530575	2118	0.25
main-p101-t2	ECBS1hc83e6a264932	ic-d4-w	101	verified	509553	2071	0.25
main-p149-t0	ECBS1h115fbf5af388	ic-d4-w	149	verified	209554	3603	0.42
main-p149-t1	ECBS1h90a74928121f	ic-d4-w	149	verified	225825	3874	0.45
main-p149-t2	ECBS1hd04149f28196	ic-d4-w	149	verified	203280	3398	0.47
main-p197-t0	ECBS1h9633c4c38036	ic-d4-w	197	verified	154962	4571	0.54
main-p197-t1	ECBS1hd7e6a1231d7d	ic-d4-w	197	verified	122383	3467	0.44
main-p29-t0	ECBS1haa8537be5e1d	ic-d4-w	29	verified	1549971	552	0.06
main-p29-t1	ECBS1h6496916a1557	ic-d4-w	29	verified	1553610	519	0.06
main-p293-t0	ECBS1hc78cfddf9b92	ic-d4-w	293	verified	329087	13740	1.75
main-p293-t1	ECBS1h5781b96b077a	ic-d4-w	293	verified	199811	8220	1.05
main-p293-t2	ECBS1h8d1fbccfafa1	ic-d4-w	293	verified	290766	11881	1.75
main-p37-t0	ECBS1h0d05ba4c3421	ic-d4-w	37	verified	3531625	2838	0.32
main-p37-t1	ECBS1h46cdd5fca268	ic-d4-w	37	verified	2604753	2117	0.24
main-p401-t0	ECBS1h314ee3e4d1b6	ic-d4-w	401	verified	110185	11460	2.23
main-p401-t1	ECBS1h090397f11941	ic-d4-w	401	verified	109045	11907	2.43
main-p509-t0	ECBS1h81d7b143eb33	ic-d4-w	509	verified	79182	12634	2.46
main-p509-t1	ECBS1h1776f19fa6c1	ic-d4-w	509	verified	67980	10567	2.27
main-p53-t0	ECBS1h4d82bfb5ff7e	ic-d4-w	53	verified	1679553	2774	0.32
main-p53-t1	ECBS1h015925fe2fbb	ic-d4-w	53	verified	2490306	4059	0.47
main-p61-t0	ECBS1hc95d970f0c12	ic-d4-w	61	verified	602126	950	0.1
main-p61-t1	ECBS1h75ab119f7681	ic-d4-w	61	verified	1076054	1628	0.18
main-p701-t0	ECBS1h91593b9e3baf	ic-d4-w	701	verified	97033	20468	3.97
main-p701-t1	ECBS1h5de606f542d8	ic-d4-w	701	verified	86109	19663	3.5

Rho-only runs

16 per rung, all verified; per-run rows in `runs.tsv` (rows whose arm is `rho-neg` and job `rho-p…`). Rung means of `S` over all distinct rho runs: 1.62, 1.39, 1.36, 1.13, 1.13, 1.05, 0.80, 0.71, 0.91, 0.93, 0.95, 1.02 for `p = 29 … 1009` (18–19 runs a rung).

Raw records

`sessions/<job>/records.jsonl` (plan, host and session files beside them); audit receipts `audits/<job>.audit.json`, list `audits/AUDITS.txt`.